

The Daily Engineer

"All the Commits Fit to Ship" — Reliability, Latency & the Quiet Pager

BENGALURU EDITION • APR 2026

CANDIDATE: PRAKHAR SHEKHAR PARTHASARTHI

FULL-STACK SWE

★ BREAKING • CAREER DESK ★

Ex-Amazon Engineer Ships Four Microservices, Cuts Latency 47%

Three years on the Travel desk: idempotent event-driven systems, Java-CDK infrastructure, and an on-call queue brought from 20+ tickets a shift to roughly five.

BY THE NEWSROOM • JAVA • SPRING BOOT • AWS • REACT • DISTRIBUTED SYSTEMS

47%

P90 LATENCY CUT

+33%

CONVERSION LIFT

195k

REDIS GET RPS

20→5

ON-CALL / SHIFT

The Lead Story

PROFILE • FROM THE CAREER DESK

A full-stack software engineer with 3+ years at Amazon across Payments and Travel, the candidate owned 4 customer-facing microservices end-to-end — design, build, deploy, and on-call.

The backend is Java-heavy: Java 17 / Spring Boot, JVM concurrency with CompletableFuture and bounded ExecutorService, idempotent event-driven architecture over SQS + Step Functions, and typed infrastructure authored entirely in AWS CDK (Java) — paired with React (SSR & CSR) + Redux on customer-facing flows.

CAREER • EXPERIENCE

Software Development Engineer

AMAZON • TRAVEL (HOTELS & FLIGHTS) • JUL 2022 – SEP 2025 • BENGALURU

- Owned 4 microservices (search, listing, room, review) on Spring Boot/MVC — REST + @Repository DynamoDB DAOs via Spring DI.
- JVM concurrency: CompletableFuture.allOf() on a bounded pool; 6 partner APIs parallelised — p90 1.1s→0.59s (-47%), conversion +33%.
- Event-driven: idempotent flight-schedule webhook — SQS + DynamoDB dedup + audit; Step Functions state machine, -43 bps CPT.
- IaC in AWS CDK (Java); migrated 2 services off CloudFormation — halved rollbacks.
- On-call mined to 20+→5/shift; 150+ incidents; 80%+ coverage.

SDE, Intern

AMAZON • 2021 • 2022 • BENGALURU

- Migrated GraphQL resolvers → Java (API Gateway + Lambda); shipped One-Click Renewal (async Java); Guice DI, 90%+ coverage.

CLASSIFIEDS • PROJECTS

Redis Server in Java

github.com/praxstack/redis-server-java

RESP2 Redis built from scratch — bounded ExecutorService, lock-free reads, hybrid TTL eviction, TCP integration tests.

SET 154k • GET 195k • INCR 128k rps • p50 < 0.3ms • 42 tests

Markdown Viewer App

github.com/praxstack/markdown-viewer-app

Real-time markdown viewer — preview, syntax highlight, Mermaid, 10 themes, PDF/HTML export. Vanilla JS, sub-100KB.

155+ Vitest tests • Mermaid • export

RECORDS • EDUCATION

M.C.A., Computer Science

MNNIT Prayagraj GPA 9.25 • 2019–2022

B.Sc. (Hons), CS

University of Delhi • 2015–2018

Honors: Amazon Pay Merchants Categories Champion (2023) — 100+ trust-buster bugs.

The Technical Review

Full Stack, Tooling & the Engineering Method — Continued

★ SECTION B • THE TOOLBENCH ★

The Complete Stack, On the Record

JAVA & JVM

Java 17, Spring Boot, Spring MVC, Spring DI, Google Guice, JUnit 5, Mockito, SLF4J, Lombok, Maven, Gradle. Concurrency: `CompletableFuture`, `ExecutorService`, `ConcurrentHashMap`, thread-pool tuning, JVM profiling, GC analysis.

DISTRIBUTED SYSTEMS

Microservices, REST, GraphQL, event-driven architecture, idempotency & dedup keys, state machines, retry with exponential backoff, DLQ handling, at-least-once semantics, TDD.

AWS

Lambda (Java), DynamoDB, SQS, Step Functions, API Gateway, AppSync, CloudWatch, ECS, CDK (Java) — typed IaC for all service infrastructure.

FRONT-END

React.js (SSR & CSR), Redux (1.5 yrs production) on customer-facing Amazon flows; HTML / CSS, JavaScript / TypeScript.

DATA & STORAGE

DynamoDB, Redis, NoSQL modelling, SQL, caching strategies.

OTHER LANGUAGES

Python, C++, VTL.

TOOLING & PROCESS

Git, GitHub, CI/CD, Gradle JaCoCo coverage gates, Agile / Scrum, on-call, code review, incident response.

EDITORIAL • THE METHOD

"The shift you don't notice is the best one." The candidate's record favors **idempotency over "just retry it,"** writes the runbook before the Slack update, and treats the on-call corpus as a backlog to automate — reclaiming roughly 2 SDE-weeks per week.

Additional dispatches on the record: GZIP on 150-200 MB catalog payloads cut wire size 95%, eliminating network-time-out failures; an auto-insurance purchase flow lifted renewals 60%+.

OP-ED • AVAILABILITY

Open to Senior SWE roles. Java / distributed systems / AWS. Reliability-first, ownership-minded, ships quiet.

"Refactored sequential downstream calls into `CompletableFuture.allOf()` — hotel landing p90 fell from 1.1s to 0.59s."

TIMELINE • ON THE RECORD

- 2025

Closed out Travel tenure — 4 services, 150+ incidents resolved.
- 2024

Idempotent flight-webhook shipped; -43 bps CPT.
- 2023

Concurrency refactor: -47% p90, +33% conversion · Merchants Champion.
- 2022

Promoted to SDE; owned hotel booking microservices.
- 2021

SDE Intern — GraphQL→Java migration.
- 2019

Began M.C.A. at MNNIT Prayagraj (GPA 9.25).